

Complete AI Summary

Beginner to Practical AI — Study Notes

A structured summary covering Artificial Intelligence, Machine Learning, Deep Learning, Generative AI, LLMs, RAG, AI Agents

Prepared as a beginner-friendly revision guide

Page 1

1. Artificial Intelligence (AI)

What is AI?

Artificial Intelligence is the field of building computer systems that can perform tasks that normally require human intelligence, such as

Main areas of AI

Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision, Robotics, Expert Systems, Speech Recognition

Types of AI

Narrow AI is designed for specific tasks. General AI refers to a hypothetical system with broad human-level intelligence. Superintelligence is a hypothetical

2. Machine Learning (ML)

Definition

Machine Learning allows systems to learn patterns from data and use those patterns to make predictions or decisions without being explicitly programmed to do so.

Supervised Learning

The model learns from labeled examples. Common tasks are classification and regression. Examples include spam detection, email filtering, and recommendation systems.

Unsupervised Learning

The model works with data without target labels. Common tasks include clustering and dimensionality reduction. Examples include market basket analysis and anomaly detection.

Reinforcement Learning

An agent learns through interaction with an environment using rewards and penalties. It is commonly discussed in robotics, game playing, and autonomous driving.

Basic workflow

Collect data -> clean data -> split data -> train model -> validate/test -> evaluate -> deploy -> monitor.

3. Deep Learning

Definition

Deep Learning uses multi-layer neural networks to learn complex representations from large datasets. It is a subset of machine learning.

Page 2

Neural network basics

A neural network contains input layers, hidden layers, and output layers. Neurons apply weighted inputs, a bias, and an activation function.

Important architectures

CNNs are widely used for visual data. RNNs process sequences. Transformers use attention mechanisms and are central to modern AI.

Training concepts

Important terms include loss function, optimizer, learning rate, epochs, batch size, backpropagation, overfitting, and regularization.

4. Natural Language Processing (NLP)

Definition

NLP focuses on enabling computers to process and generate human language.

Common tasks

Text classification, sentiment analysis, translation, summarization, question answering, named-entity recognition, information extraction.

Tokenization

Text is divided into tokens that a language model can process. A token may be a word, part of a word, punctuation mark, or an embedding.

Important idea

Modern NLP systems commonly represent language using learned numerical representations called embeddings.

5. Generative AI

Definition

Generative AI creates new content such as text, images, audio, video, or code based on learned patterns.

Examples

Text generation, image generation, code assistants, document summarization, and conversational systems.

Prompting

A good prompt clearly states the task, context, constraints, expected format, and important input data.

Limitations

Generative models can produce incorrect information, misunderstand ambiguous prompts, or generate plausible but unsupported content.

6. Large Language Models (LLMs)

Page 3

What is an LLM?

An LLM is a neural network trained on large amounts of text to model language and generate responses.

Transformer

Transformers use attention to determine which parts of an input are relevant to other parts. This architecture made large-scale

Inference

Inference is the process of using a trained model to produce an output for a new input.

Context window

The context window is the amount of information a model can consider in one interaction. Larger context does not automatically

7. Embeddings and Vector Search

Embeddings

An embedding converts text or another object into a numerical vector that captures useful semantic information.

Similarity

Vectors can be compared using measures such as cosine similarity to find semantically related content.

Vector database

A vector database stores embeddings and supports similarity search. It is useful for semantic retrieval.

Examples

Common technologies include FAISS, Chroma, Pinecone, Weaviate, Milvus, and pgvector.

8. RAG — Retrieval-Augmented Generation

Definition

RAG combines retrieval with generation. Instead of relying only on the model's internal knowledge, the system retrieves relevant

Pipeline

Documents -> load -> split into chunks -> create embeddings -> store in vector database -> retrieve relevant chunks -> send co

Why use RAG?

It is useful for private documents, frequently changing information, company knowledge bases, manuals, PDFs, and domain-sp

Key issue

Page 4

Retrieval quality strongly affects answer quality. Poor chunking, weak embeddings,

9. AI Agents

Definition

An AI agent is a system that can reason about a task, use tools, observe results, and continue until it reaches an appropriate o

Typical components

LLM + instructions + tools + memory/state + execution loop + optional retrieval.

Tools

Tools may include calculators, search, weather APIs, databases, file readers, or business APIs.

Agent loop

Understand goal -> decide action -> call tool -> inspect result -> decide next step -> produce final response.

Agent vs chatbot

A basic chatbot mainly responds to messages. An agent can take actions and use external tools as part of completing a task.

10. LangChain

Purpose

LangChain is a framework for building applications around language models, including chains, tool use, retrieval, agents, and i

Common concepts

Models, prompts, messages, tools, retrievers, vector stores, parsers, memory/state, and agents.

Typical RAG structure

Document loader -> text splitter -> embeddings -> vector store -> retriever -> prompt -> chat model -> answer.

Good practice

Keep API keys in environment variables, separate configuration from application logic, validate tool inputs, and log useful non-s

11. LangGraph

Definition

Page 5

LangGraph is designed for building stateful, multi-step agent workflows as graphs.

Graph concepts

Nodes perform work, edges define transitions, and state carries information between steps.

Why use it?

It is useful when an agent requires predictable workflows, branching, loops, persistence, human approval, or multiple specialization.

12. AI APIs

What is an API?

An Application Programming Interface lets one software system communicate with another using defined requests and responses.

LLM API flow

Application sends a request containing model information and messages -> API processes it -> application receives a response.

Security

Never hard-code private API keys into public source code. Store secrets in environment variables or a secure secret manager.

13. Python for AI

Important skills

Variables, functions, classes, lists, dictionaries, tuples, exceptions, file handling, modules, virtual environments, and package management.

Useful libraries

NumPy for numerical computing, pandas for data manipulation, matplotlib for visualization, scikit-learn for classical ML, PyTorch for deep learning.

Environment

A virtual environment keeps project dependencies isolated. A requirements.txt file can record packages needed to reproduce a project.

14. Data and Model Evaluation

Data quality

AI performance depends heavily on the quality, relevance, coverage, and correctness of data.

Classification metrics

Page 6

Accuracy measures overall correctness; precision measures how many predicted p

Regression metrics

MAE measures average absolute error. MSE measures average squared error. RMSE is the square root of MSE.

LLM evaluation

Check factuality, relevance, completeness, instruction following, safety, latency, cost, and consistency. For RAG, also evaluate

15. Fine-Tuning vs RAG

Fine-tuning

Fine-tuning changes model parameters using task-specific training data. It can be useful for behavior, style, or specialized task

RAG

RAG supplies external information at inference time. It is often preferable when knowledge changes frequently or comes from p

Simple rule

If the main problem is missing or changing knowledge, consider RAG. If the main problem is behavior or task-specific adaptatio

16. AI Project Architecture

Typical application

Frontend -> backend/API -> AI orchestration -> model provider -> tools/retrieval -> data sources -> response.

Production concerns

Authentication, authorization, secret management, rate limits, validation, monitoring, error handling, caching, cost control, testin

Deployment

A project can be deployed on a cloud platform after adding the correct build/start commands, dependencies, environment varia

17. Responsible AI

Core principles

Safety, privacy, transparency, fairness, reliability, accountability, and human oversight.

Page 7

Privacy

Do not collect or expose unnecessary personal information. Protect credentials and sensitive data.

Human oversight

High-impact decisions should have appropriate human review and verification rather than blindly trusting model output.

18. Quick Revision — Important Terms

AI: Broad field of intelligent computer systems.

ML: Learning patterns from data.

Deep Learning: Machine learning using multi-layer neural networks.

NLP: Processing and generating human language.

LLM: Large language model for language understanding/generation.

Embedding: Numerical vector representation used for semantic comparison.

RAG: Retrieve external information and use it to generate an answer.

Agent: AI system that can reason and use tools to complete tasks.

API: Interface for communication between software systems.

Vector Database: Database optimized for storing and searching vector representations.

19. Beginner AI Roadmap

Step 1: Learn Python fundamentals and basic Git/GitHub.

Page 8

Step 2: Learn NumPy, pandas, data visualization, and basic statistics.

Step 3: Learn machine learning with scikit-learn.

Step 4: Learn neural networks and deep learning concepts.

Step 5: Learn NLP, transformers, prompting, and LLM APIs.

Step 6: Build a RAG PDF/document chatbot.

Step 7: Build tool-using AI agents.

Step 8: Learn LangChain/LangGraph and build structured workflows.

Step 9: Deploy projects and learn monitoring/security.

Step 10: Build a portfolio with several practical projects.

Page 9

AI Cheat Sheet

Concept | One-line meaning

AI | Machines performing tasks associated with intelligence

ML | Learning patterns from data

DL | Neural-network-based machine learning

NLP | Working with human language

LLM | Large neural language model

Embedding | Vector representation of information

RAG | Retrieve context before generating an answer

Agent | LLM-based system that can use tools/actions

API | Programmatic interface between systems

Vector DB | Stores and searches embeddings

Prompt | Instructions/context given to an AI model

Note: This PDF is a study/revision guide. AI tools, models, APIs, and frameworks change over time, so current official documents may differ.
Page 10